DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

NITK, Surathkal

Algorithms And Complexity(Code: CS700)

I Sem MTech. CSE

September 27, 2024

Registration No.

Time: 1 hour — *Quiz - 1* — Maximum Marks: 20

Note:- *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. Solve the following recurrences. Assume that $T(n)$ is constant for sufficiently small n.
   a. $T(n) = 4T(\frac{n}{2}) + n^2\sqrt{n}$
   b. $T(n) = 4T(\sqrt{n}) + log^2 n$

[02+02]

2. Use recursion tree to determine a good asymptotic upper bound on the recurrence $T(n) = 3T(\frac{n}{4}) + n$. Use the substitution method to verify your answer. [04]

3. Derive the best, worst and average case analysis of quick sort algorithm.

[06]

4. In the algorithm SELECT(the kth smallest in a set whose worst case linear time), the input elements are divided into groups of 5. Will the algorithm work in linear time, if they are divided into groups of 7? How about groups of 3? Analyze the running time of this algorithm? [06]

$T(n) = T(k-1) + T(n-k) + O(n)$

k. T

$\frac{n}{7} \cdot 4\left(\frac{n}{14}\right)$

$C = 4$

$n - \left(\frac{2n}{7} - 8\right)$

$\frac{63}{8}$

$\frac{5n}{7} + 8$

$n - \frac{2n}{6} - 4 = \frac{4n}{6} + 4 = \frac{2}{3}n + 4$

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Algorithms And Complexity (Code: CS700)

I Sem MTech. CSE

October 07, 2024

Registration No.

Time: 1:30 hour | *Mid Sem Exam* | Maximum Marks: 40

Note:- *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. If $f(n)$ is a kth degree polynomial, then show that $f(n) = O(n^k)$. [04]

2. You are given an unsorted integer array A containing n distinct integers (assume n is a power of 2). You are supposed to design an algorithm that finds the minimum and the second minimum element in the array A. The running time for this problem is measured in terms of the number of pairwise comparisons that your algorithm performs in the worst case. (Note that you can find the minimum and the second minimum using $(2n-3)$ comparisons by a linear scan.) Design a divide-and-conquer algorithm that solves the problem using at most $(\frac{3n}{2} - 2)$comparisons. [06]

3. Show how to multiply complex numbers $a + bi$ and $c + di$ using only three multiplications of real numbers. The algorithm should take a, b, c and d as input and produce the real component $ac - bd$ and the imaginary component $ad + bc$ separately. [04]

4. Let $X[1 \ldots n]$ and $Y[1 \ldots n]$ be two arrays each containing n numbers already in sorted order. Design and analyse an efficient divide and conquer $O(\log n)$-time algorithm to find the median of all $2n$ elements in arrays X and Y. [08]

5. The Discrete Knapsack Problem is defined as follows. A thief is robbing a store that has n items. The ith item is worth v_i dollars and weighs w_i pounds, where v_i and w_i are integers. He wants to take as valuable a load as possible, but he can carry at most W pounds in his knapsack, for some integer W . Which items should he take?

   In the Continuous Knapsack Problem, the setup is the same, but the thief can take fractional amounts of items, rather than having to take all or nothing of an item.

   Which of the two problems exhibit the greedy-choice property? Briefly argue why or why not with example. [08]

6. Given n files of length $m_1, m_2, \ldots\ldots\ldots m_n$, the optimal Tape storage problem is to find which order is the best to store them on the tape, assuming that
   i) each retrival starts with the tape rewound.
   ii)Each retrival takes time equal to the length of the preceeding files in the tape plus the length of the retrival file

   a) Describe a greedy algorithm for this problem.
   b) What is the running time of your algorithm?
   c) Prove your algorithm is correct. [10]

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Algorithms And Complexity (Code: CS700)

November 22, 2024

1 Semester M.Tech CSE

Registration No.

Time: 60 minutes. *Quiz-2* Maximum Marks: 20

Note:- *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. Given two strings $x = x_1x_2x_3...x_n$ and $y = y_1y_2....y_m$, a longest common subsequence is the largest k such that there exist indices $i_1 < i_2 < ..... < i_k$ and $j_1 < j_2 < ..... < j_k$ with $x_{i_1}x_{i_2}.....x_{i_k} = y_{j_1}y_{j_2}.....y_{j_k}$. Obtain an $O(mn)$ algorithm to find the length of the longest common subsequence in the given two input strings. (Notice that we need a subsequence here and not a substring, that is, the indices do not have to be continuous.)

[08]

2. Explain Polynomial time reducibility.

[02]

3. Define Class P, Class NP, Class NPC and Class NP-Hard problems.

[04]

4. Give a polynomial time reduction from 3-SAT to Vertex Cover problem. Prove that Vertex Cover is NP-Hard.

[06]

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Algorithms And Complexity (Code: CS700)

November 27, 2024

I Semester M.Tech CSE

Registration No.

Time: 3Hrs. *End Semester Exam* Maximum Marks: 80

Note:- *Answer all the questions. Missing data may be suitably assumed, provided the assumption does not change the difficulty of the problem.*

1. Suppose you have k sorted arrays, each with n elements, and you want to combine them into a single sorted array of kn elements.

   i) Here's one strategy: Using the merge procedure , merge the first two arrays, then merge in the third, then merge in the fourth, and so on. What is the time complexity of this algorithm, in terms of k and n?

   ii) Give a more efficient solution to this problem, using divide-and-conquer.

   [08]

2. Assume that the numbers given below represent counts of letters in the hundreds from a file. For example, in the file there will be exactly 20 * 100 occurrences of the letter 'a' , 11*100 occurrence of the letter 'c', etc. a: 20, c:11, d:2, e: 10, o:15, m:8, s:10, t:22, u: 2

   a) What is an optimal Huffman code based on the following set of frequencies?

   1) Draw the tree. Show your work at every step.

   2) Fill in the table the Huffman encoding for each letter.

   3) We encode the file using the Huffman codes produced above. How much memory will the file require with this encoding?

   b) Fixed-length encoding:

   1) Fill in the table the fixed-length encoding for each letter.

   2) What will be the file size when the fixed-length encoding is used?

   [12]

3. Let U and V be two strings of character. We want to transform U into V with the smallest possible number of operations of the following type

   1)Delete a character

   2)Add a character

   3)change a character

   Write a dynamic programming algorithm that finds the minimum number of operations needed to transform U into V, and that tells us what these operations are.

   How much time does your algorithm takes as a function of the lengths of U and V. [12]

4. Construct a maximum length common subsequence from the completed C table and the original sequences $X =< A, B, C, B, D, A, B >$ and $Y =< B, D, C, A, B, A >$ [07]

5. Give a polynomial time reduction from CNF to 3-SAT problem.

   [07]

CSE,NITK Surathkal- I Sem- MTech.-2023-2024- Mid-Sem Exam-Oct 2023
Algorithms and Complexity(CS700)

Notation: n-number of vertices, m- number of edges.

1. Let G be a forest on n vertices. If the number of trees in the forest G is k, then the number of edges of G is ——. (1)

2. Let e be an edge with unique maximum weight in G having more than $n-1$ edges. Then e is not part of minimum spanning tree in G. TRUE/FALSE. Justify your answer. (1)

3. If T is a tree having 'l' leaves, then prove that the number of nodes with degree at least three are at most $l-2$. (3)

4. Given a directed graph G and a vertex v in G, give an efficient algorithm to check if the vertex v is part of a directed cycle in the given directed graph G. Prove the correctness of your algorithm. (3)

5. Given a set of n intervals on a real line, give an efficient algorithm to find the minimum number of colors required to color the intervals in such a way that if two intervals intersect, they are assigned different colors. Prove the correctness of your algorithm. (4)

6. Let G be an edge weighted graph and let $U \subseteq V(G)$. Give an algorithm to find minimum weight spanning tree T with the property that the nodes in U are leaves of T. i.e., $U \subseteq L(T)$, where $L(T)$ is a set of leaves of T. (Note that this tree T need not necessarily be a minimum spanning tree in G.) Prove the correctness of your algorithm. Also give its run time complexity. (4)

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I Sem- MTech.-2023-2024- Final Exam-December
Algorithms and Complexity (CS700) - Total Marks : 40

1. (a) We can use BFS algorithm from source s to get single source shortest paths from s to all other vertices in an unweighted graph. TRUE/FALSE. (1)

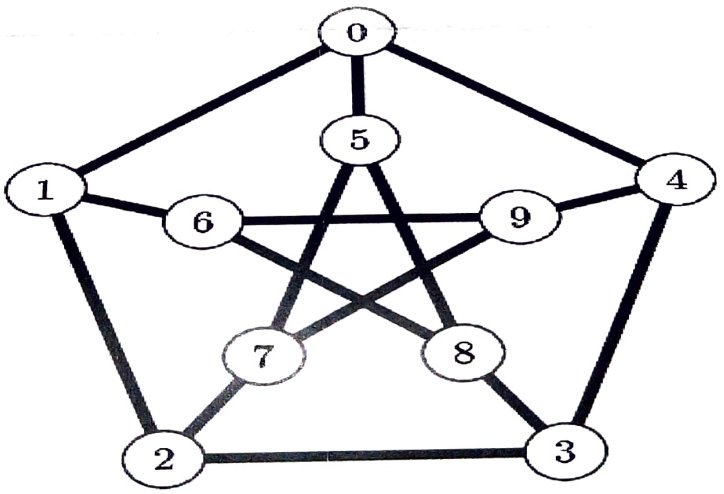
   (b) Hamiltonian path problem in a DAG can be solved in polynomial time. TRUE/False (1)

   (c) Let G be a connected graph with each edge having unique weight. Then the number of minimum spanning trees in G is ..... (1)

   (d) If you are doing a bfs traversal on a complete graph, then the number of levels in the bfs tree associated with the traversal is ..... (1)

2. The shortest path between two vertices is part of some MST. TRUE/FALSE. Justify your answer. (2)

3. Does the following graph have a Hamiltonian cycle? If YES, output the cycle. If NO, output a longest cycle in the graph? Also give maximum matching and shortest cycle in the graph. (4)



4. Prove that a tree on n vertices has $n-1$ edges. (2)

5. Prove that if e is an edge of a minimum spanning tree of graph G, then it is a minimum weight edge across some cut S in the graph G. (3)

6. Assuming $3-SAT$ to be NP-Complete, prove that the maximum independent set problem is NP-Complete. (5)